

HCSM-13

The Processor T and the Fermion/Boson-Class Correspondence The Rank-Ten Projection, the Fermion/Boson Split, and the Localization of Chirality Breaking

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus. The fourteen modes at the self-paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. On this multiplet the processor T is the rank-ten orthogonal projection that separates the ten Standard Model fermions from the four Standard Model bosons. This paper proves two theorems about the processor and its action.

First, the *Fermion/Boson-Class Correspondence Theorem*: the processor commutes with the class- C projector of the winding partition, its image is exactly the union of winding classes A and B , and its kernel is exactly winding class C . Equivalently, $T = I - P_C$. The fermion/boson split is the coarse-graining of the winding class partition that identifies classes A and B .

Second, the *Chirality-Breaking Localization Theorem*: the commutator of the processor with the chiral reflection is supported exactly on the two-element set consisting of the dark photon and the tau. On the two-dimensional plane spanned by these two modes, the commutator is a fourth-order rotation generator. On the other twelve modes, the processor and the chiral reflection commute.

All theorems are HCSM-native, and all numerical claims are verified at machine precision.

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1 Introduction

1.1 The processor in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus [1]. The fourteen modes at the self-paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation [3, 9]. Their dihedral representation theory determines the gauge algebra [5]; their mass-ordering labels determine the mass spectrum [13]; the zero-state vector from the 14×19 measurement matrix determines the flavor structure [8].

Above the substrate, the *processor* T reads the fourteen-mode multiplet and separates it into a fermion sector and a boson sector. The processor is a rank-ten orthogonal projection: it keeps ten modes (the Standard Model fermions) and kills four modes (the Standard Model bosons). Its matrix is diagonal in the mode basis.

This paper proves that the processor's output is not an independent structure on the multiplet. It is the coarse-graining of the winding class partition that identifies two of the three winding classes. And the failure of the processor to commute with the framework's chiral reflection is localized to a single two-mode plane.

1.2 One-line summary

The processor T is exactly the complement of the class- C projector. The fermion/boson split is the same as the winding class partition, with classes A and B fused into a single fermion sector. Chirality is broken only on the plane spanned by the dark photon and the tau: twelve of the fourteen modes see no chirality breaking at all.

1.3 The two theorems

Theorem 1.1 (Fermion/Boson-Class Correspondence). *The processor T commutes with the class- C projector P_C of the winding partition and equals*

$$T = I - P_C = P_A + P_B. \quad (1)$$

Equivalently, $\text{im}(T) = A \cup B$ and $\ker(T) = C$.

Theorem 1.2 (Chirality-Breaking Localization). *The commutator $[T, P_{rs}]$ is supported exactly on the two-element set $\{\text{dark}, \tau\}$. On the two-dimensional plane spanned by these modes, the commutator is a fourth-order rotation generator. On the other twelve modes, T and P_{rs} commute.*

1.4 Cross-references to the HCSM paper stack

This paper depends on three companion papers in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-04** [3]: the fourteen-mode multiplet and its dihedral orbit decomposition.
- **HCSM-14** [7]: the winding generator spectrum and the winding class partition.
- **HCSM-26** [10]: the observer pair, which supplies the chiral reflection and the fixed-locus theorem.

Additional references to HCSM-22 (the zero state, [8]) and HCSM-27 (Klein orbit structure, [11]) appear where needed.

1.5 Organization

Section 2 states the setup: the fourteen-mode multiplet, the winding class partition, the processor T , and the chiral reflection. Section 3 proves the Fermion/Boson-Class Correspondence Theorem. Section 4 proves the Chirality-Breaking Localization Theorem. Section 5 discusses the structural interpretation. Section 6 gives reproducibility notes. Section 7 concludes.

2 Setup

All statements in this section are HCSM-native; the reader is referred to HCSM-00 [1] for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad |\Lambda| = N = 64, \quad (2)$$

selected by the unified selection theorem of HCSM-00 [1] (equal entropy spacing and maximal winding cancellation). The discrete five-point Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (3)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. The unique self-paired eigenvalue is $\lambda = 4$, with multiplicity $2(L - 1) = 14$.

In the standard mode order used throughout HCSM, matching the zero-state vector of HCSM-22 [8], the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (4)$$

The mode coordinates (n_1, n_2) in this order are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (5)$$

The fourteen modes decompose under the dihedral point group D_4 of the square into three orbits of sizes $(2, 8, 4)$ [3, 4].

2.2 The winding generator and class partition

The winding generator on the fourteen-mode multiplet is the Hermitian operator

$$H_{\text{wind}} = i \sin\left(\frac{\pi}{8} M\right), \quad M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}, \quad (6)$$

with the sine applied elementwise. Its spectrum is [7]

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (7)$$

The fourteen modes partition into three *winding classes* by frequency:

$$A = \{\nu, e, t\}, \quad |A| = 3, \quad (8)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad |B| = 7, \quad (9)$$

$$C = \{\text{dark}, W, Z, H\}, \quad |C| = 4. \quad (10)$$

The class projectors P_A, P_B, P_C are diagonal in the mode basis, mutually orthogonal, idempotent, and sum to the identity:

$$P_A P_B = P_B P_C = P_A P_C = 0, \quad P_A + P_B + P_C = I_{14}. \quad (11)$$

2.3 The processor T

Definition 2.1 (Processor). The *processor* T is the rank-ten orthogonal projection onto the fermion sector, defined as the sum

$$T = P_{\text{mesh}} + P_{\tau}, \quad (12)$$

where P_{mesh} projects onto the nine mesh resonances (those modes with residual $\delta \neq 0$ as defined in HCSM-11 [6]) and P_{τ} projects onto the unique mode with zero-state amplitude $s = +1$ [8].

In the standard mode order (4), the processor is diagonal with entries

$$\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1). \quad (13)$$

The fermion sector and the boson sector are therefore

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad B_{\text{bos}} = \{\text{dark}, W, Z, H\}, \quad |B_{\text{bos}}| = 4. \quad (14)$$

Proposition 2.2 (Processor properties). *The processor satisfies $T^2 = T$, $T^\dagger = T$, and $\text{rank}(T) = 10$.*

Proof. Orthogonal projection by construction. Verified at machine precision in the companion verification script. $\square$

2.4 The chiral reflection rs

Definition 2.3 (Chiral reflection). The *chiral reflection* is the map

$$rs : (n_1, n_2) \longmapsto (n_2, n_1) \pmod{8}. \quad (15)$$

It is an element of the dihedral point group D_4 , and it satisfies $rs^2 = e$.

The map rs was identified in HCSM-26 [10] as the unique chiral reflection of the dihedral group: the unique element that fixes exactly the coordinate pair $\{(2, 2), (6, 6)\}$ on the fourteen-mode multiplet and anticommutes with the winding generator.

Let P_{rs} denote the 14×14 permutation matrix implementing rs in the mode basis. It satisfies

$$P_{rs}^2 = I_{14}, \quad P_{rs}^\dagger = P_{rs}. \quad (16)$$

3 The Fermion/Boson-Class Correspondence

This section proves that the processor T is exactly the complement of the class- C projector.

3.1 The complement formulation

Theorem 3.1 (Fermion/Boson-Class Correspondence). *The processor T commutes with the class- C projector P_C and equals*

$$T = I_{14} - P_C = P_A + P_B. \quad (17)$$

Equivalently:

- (i) $\text{im}(T) = A \cup B = F$;
- (ii) $\text{ker}(T) = C = B_{\text{bos}}$;
- (iii) $[T, P_C] = 0$.

Proof. Direct inspection of the diagonal entries of T and of the class membership of each mode. The processor T is diagonal, with $\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$ in the standard mode order. Hence:

- The image of T (modes with diagonal entry 1) is

$$\text{im}(T) = \{\nu, e, u, d, s, \mu, c, \tau, b, t\} = F. \quad (18)$$

- The kernel of T (modes with diagonal entry 0) is

$$\text{ker}(T) = \{\text{dark}, W, Z, H\} = B_{\text{bos}}. \quad (19)$$

On the other hand, the winding classes are given by (8)–(10). Hence

$$A \cup B = \{\nu, e, u, d, s, \mu, c, \tau, b, t\} = F = \text{im}(T), \quad (20)$$

$$C = \{\text{dark}, W, Z, H\} = B_{\text{bos}} = \text{ker}(T). \quad (21)$$

Since T is diagonal and P_C is diagonal, they commute. The identity $T = I_{14} - P_C$ is verified by comparing

$$\text{diag}(I_{14} - P_C) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1) \quad (22)$$

with $\text{diag}(T)$.

The full matrix identity is verified at machine precision:

$$\|T - (I_{14} - P_C)\|_F = 0. \quad (23)$$

□

Corollary 3.2 (The fermion/boson split is a coarse-graining). *The fermion/boson split is the coarse-graining of the winding class partition that identifies classes A and B . Equivalently, the fermion sector is the image of the processor and the boson sector is the kernel of the processor.*

Proof. Immediate from Theorem 3.1. □

Remark 3.3. Theorem 3.1 states that the processor does not add structure beyond the winding class partition. The winding class partition is finer-grained: it separates class A from class B . The fermion/boson split ignores that distinction. The winding class partition and the fermion/boson split encode the same physical content, viewed at different coarse-graining scales.

3.2 The Landauer channel

Proposition 3.4 (Landauer entropy reduction). *The processor reduces the entropy of the maximally mixed state on the 56-dimensional Hodge sector to the entropy of the maximally mixed state on the fermion/boson product. The entropy reduction is*

$$\Delta S = \log 56 - (\log 10 + \log 4) = \log 56 - \log 40 = \log\left(\frac{7}{5}\right) \approx 0.33647 \text{ nats}. \quad (24)$$

This saturates the deterministic Landauer bound.

Proof. Direct computation. The 56-dimensional sector is the Hodge lift of the 14-mode multiplet at $\lambda = 4$ (HCSM-02 [2]); the fermion/boson product has dimension $10 \times 4 = 40$; the ratio $56/40 = 14/10 = 7/5$ yields the stated entropy reduction. □

4 The Chirality-Breaking Localization Theorem

This section proves that the processor T and the chiral reflection rs fail to commute only on the two-dimensional plane spanned by the dark photon and the tau.

4.1 The commutator

Theorem 4.1 (Chirality-Breaking Localization). *The commutator of the processor with the chiral reflection,*

$$C_{rs} := T \cdot P_{rs} - P_{rs} \cdot T, \quad (25)$$

is supported exactly on the two-element set

$$\{\text{dark}, \tau\}. \quad (26)$$

On every other pair of indices the commutator vanishes. The two-dimensional plane spanned by the dark photon and the tau is the framework's unique chirality-breaking plane.

Proof. Direct computation of the commutator. The processor T is diagonal with

$$\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1), \quad (27)$$

and the chiral reflection P_{rs} permutes the modes by the diagonal reflection. We compute

$$(T \cdot P_{rs})_{ij} = T_{ii} (P_{rs})_{ij}, \quad (P_{rs} \cdot T)_{ij} = (P_{rs})_{ij} T_{jj}. \quad (28)$$

For the commutator to be nonzero at entry (i, j) we require both $(P_{rs})_{ij} = 1$ (so that rs maps mode j to mode i) and $T_{ii} \neq T_{jj}$.

The chiral reflection maps each mode as follows in the standard mode order:

$$rs(\nu) = e, \quad rs(e) = \nu, \quad (29)$$

$$rs(u) = s, \quad rs(s) = u, \quad (30)$$

$$rs(d) = b, \quad rs(b) = d, \quad (31)$$

$$rs(\mu) = c, \quad rs(c) = \mu, \quad (32)$$

$$rs(\tau) = \text{dark}, \quad rs(\text{dark}) = \tau, \quad (33)$$

$$rs(W) = W \text{ (fixed)}, \quad rs(Z) = H, \quad (34)$$

$$rs(H) = Z, \quad rs(t) = t \text{ (fixed)}. \quad (35)$$

For the pairs where rs swaps two modes we examine whether their diagonal entries in T differ:

$$\nu, e : \quad T_{\nu\nu} = 1, \quad T_{ee} = 1 \quad (\text{equal}), \quad (36)$$

$$u, s : \quad T_{uu} = 1, \quad T_{ss} = 1 \quad (\text{equal}), \quad (37)$$

$$d, b : \quad T_{dd} = 1, \quad T_{bb} = 1 \quad (\text{equal}), \quad (38)$$

$$\mu, c : \quad T_{\mu\mu} = 1, \quad T_{cc} = 1 \quad (\text{equal}), \quad (39)$$

$$\tau, \text{dark} : \quad T_{\tau\tau} = 1, \quad T_{\text{dark}, \text{dark}} = 0 \quad (\text{different}), \quad (40)$$

$$Z, H : \quad T_{ZZ} = 0, \quad T_{HH} = 0 \quad (\text{equal}). \quad (41)$$

The only pair with a nonzero commutator is $\{\text{dark}, \tau\}$. The two fixed points W and t contribute nothing, since rs acts trivially on them and hence cannot create off-diagonal commutator entries. Therefore the commutator is supported exactly on the pair $\{\text{dark}, \tau\}$. $\square$

4.2 The fourth-order structure of the commutator

Proposition 4.2 (The commutator as a fourth-order rotation generator). *On the two-dimensional plane spanned by the dark photon and the tau, the commutator is*

$$C_{rs}|_{\{\text{dark}, \tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}. \quad (42)$$

Its square is the negative of the plane projector and its fourth power is the positive plane projector:

$$C_{rs}^2|_{\{\text{dark}, \tau\}} = -P_{\{\text{dark}, \tau\}}, \quad C_{rs}^4|_{\{\text{dark}, \tau\}} = +P_{\{\text{dark}, \tau\}}. \quad (43)$$

Hence C_{rs} generates a fourth-order rotation on the chirality-breaking plane.

Proof. Direct computation. The diagonal entries of C_{rs} on the plane vanish:

$$(C_{rs})_{\tau\tau} = T_{\tau\tau} - T_{\tau\tau} = 0, \quad (C_{rs})_{\text{dark,dark}} = T_{\text{dark,dark}} - T_{\text{dark,dark}} = 0. \quad (44)$$

The off-diagonal entries are

$$(C_{rs})_{\tau,\text{dark}} = -(P_{rs})_{\tau,\text{dark}} T_{\text{dark,dark}} = -1, \quad (C_{rs})_{\text{dark},\tau} = +(P_{rs})_{\text{dark},\tau} T_{\tau\tau} = +1. \quad (45)$$

Hence

$$C_{rs}|_{\{\text{dark},\tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}. \quad (46)$$

Squaring gives

$$C_{rs}^2|_{\{\text{dark},\tau\}} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -P_{\{\text{dark},\tau\}}, \quad (47)$$

and fourth power gives

$$C_{rs}^4|_{\{\text{dark},\tau\}} = \begin{pmatrix} +1 & 0 \\ 0 & +1 \end{pmatrix} = +P_{\{\text{dark},\tau\}}. \quad (48)$$

The commutator therefore generates a fourth-order rotation on the chirality-breaking plane, verified at machine precision in the companion verification script. $\square$

4.3 Commutation on the other twelve modes

Corollary 4.3 (Commutation elsewhere). *The processor T and the chiral reflection rs commute on the twelve modes not in the chirality-breaking plane $\{\text{dark},\tau\}$.*

Proof. Immediate from Theorem 4.1. The restriction of the commutator to the subspace spanned by the other twelve modes is identically zero. $\square$

Remark 4.4. The localization of chirality breaking is sharp. On the twelve modes outside the plane $\{\text{dark},\tau\}$, the processor preserves the chiral reflection's action exactly. Only on the two-mode plane does chirality break, and the breaking is a fourth-order rotation. This makes the framework's chirality content highly constrained.

5 Discussion

5.1 What the processor is

The processor T is not an independent structure on the fourteen-mode multiplet. It is the complement of the class- C projector. Its image is the union of classes A and B ; its kernel is class C . The fermion/boson split is exactly the coarse-graining of the winding class partition that fuses classes A and B .

This means that the processor's output is determined by the winding class partition, which in turn is determined by the dihedral orbit partition plus the fermion/boson homogeneity constraint of HCSM-28 [12]. The whole processor structure is therefore determined by the substrate, without independent input.

5.2 What the chirality breaking is

The chirality breaking of the processor is not a global feature. It is localized to the plane $\{\text{dark}, \tau\}$. On the twelve other modes, the processor and the chiral reflection commute exactly.

The two chirality-breaking modes are the dark photon (a boson in the kernel of T) and the tau (a fermion in the image of T). Together they form the unique two-dimensional subspace on which the processor breaks the symmetry.

This two-mode plane is the intersection of the processor with the chirality structure: it is the unique place where the processor's fermion/boson distinction conflicts with the dihedral reflection.

5.3 Relation to the observer pair

The observer pair of HCSM-26 [10] is the pair of fixed points of the chiral reflection: the W boson and the top quark. These are precisely the two modes on which the chiral reflection acts trivially. Since T and rs commute on W and t , the observer pair is not part of the chirality-breaking plane.

The framework's structure therefore separates cleanly: the observer pair is the fixed locus of the chiral reflection, and the chirality-breaking plane is the unique two-mode plane where the processor and the chiral reflection disagree. These are two independent structural features of the fourteen-mode multiplet.

5.4 Relation to the Klein orbit structure

The two chirality-breaking modes, the dark photon and the tau, are members of the four-cycle orbit of the Klein group (HCSM-27 [11]). Specifically, the four-cycle orbit is

$$\{\text{dark}, s, u, \tau\}, \quad (49)$$

and the chirality-breaking pair is the subset $\{\text{dark}, \tau\}$. The chiral reflection and the half-period translation both individually swap these two modes with other modes in the four-cycle; the product fixes them together, which is why the four-cycle orbit has trivial pointwise stabilizer.

6 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion verification script is provided in the HCSM companion repository at the verification folder with the number thirteen. The script constructs the fourteen-mode multiplet, the winding class projectors, the processor T , and the chiral reflection P_{rs} from HCSM-native definitions and verifies:

1. The processor properties $T^2 = T$, $T^\dagger = T$, $\text{rank}(T) = 10$.
2. The matrix identity $T = I_{14} - P_C$ to machine precision.
3. The image $\text{im}(T) = A \cup B$ and the kernel $\text{ker}(T) = C$.
4. The commutation $[T, P_C] = 0$.
5. The commutator $[T, P_{rs}]$ restricted to the pair $\{\text{dark}, \tau\}$ and zero on the other twelve modes.
6. The fourth-order structure: $[T, P_{rs}]^2 = -P_{\{\text{dark}, \tau\}}$ and $[T, P_{rs}]^4 = +P_{\{\text{dark}, \tau\}}$.

7. The Landauer entropy reduction $\Delta S = \log(7/5)$.

Running the master verification script `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

7 Conclusion

This paper has proven two theorems about the processor T of the Hodge Complex Standard Model.

The first is the *Fermion/Boson-Class Correspondence Theorem* (Theorem 3.1): the processor is exactly the complement of the class- C projector of the winding partition. Its image is the union of winding classes A and B ; its kernel is class C . The fermion/boson split is the coarse-graining of the winding class partition that fuses classes A and B . Equivalently, $T = I_{14} - P_C$.

The second is the *Chirality-Breaking Localization Theorem* (Theorem 4.1): the processor's commutator with the chiral reflection is supported exactly on the two-mode plane spanned by the dark photon and the tau. On this plane the commutator is a fourth-order rotation generator. On the other twelve modes the processor and the chiral reflection commute exactly.

The framework's chirality content is therefore highly constrained. The chirality breaking is localized to a single two-mode plane, and the observer pair of HCSM-26 (the W boson and the top quark) is exactly the complement: the fixed locus of the chiral reflection. The processor's output and its chirality breaking are both determined by the winding class partition and the dihedral reflection, with no independent input.

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